

Computer Architecture :

It deals with instructions, addressing modes, ALU, pipelining etc (Internal Design)

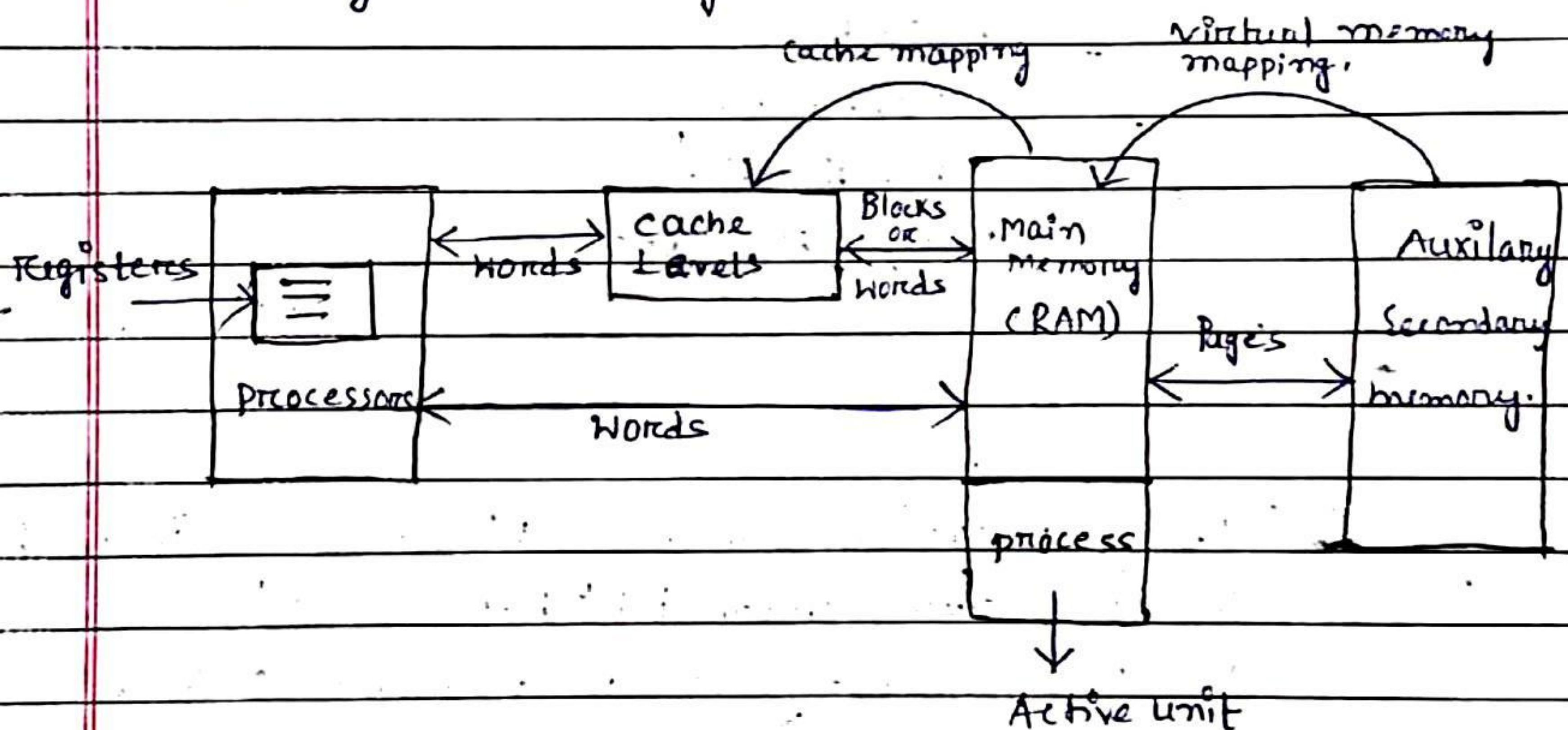
Computer Organization :

It deals with how various memory and I/O interact with a system.

Computer design :

It deals with hardware design.

• Memory Interfacing -

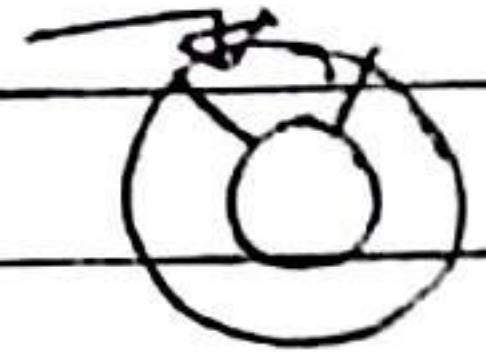


• Memory Hierarchy :

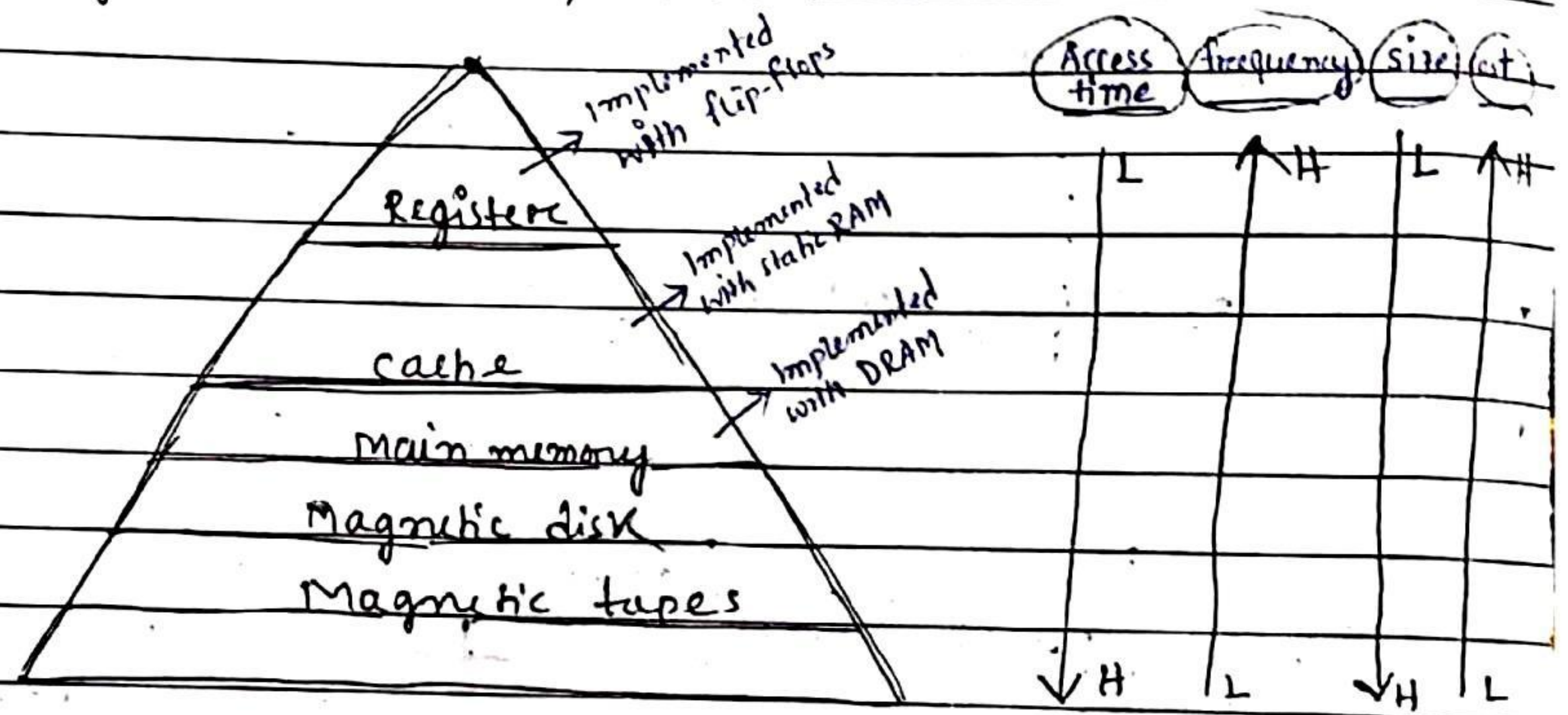
cache levels }
Main memory } Random Access.



Magnetic disk → semi Random Access.



Magnetic tapes → sequential Access.



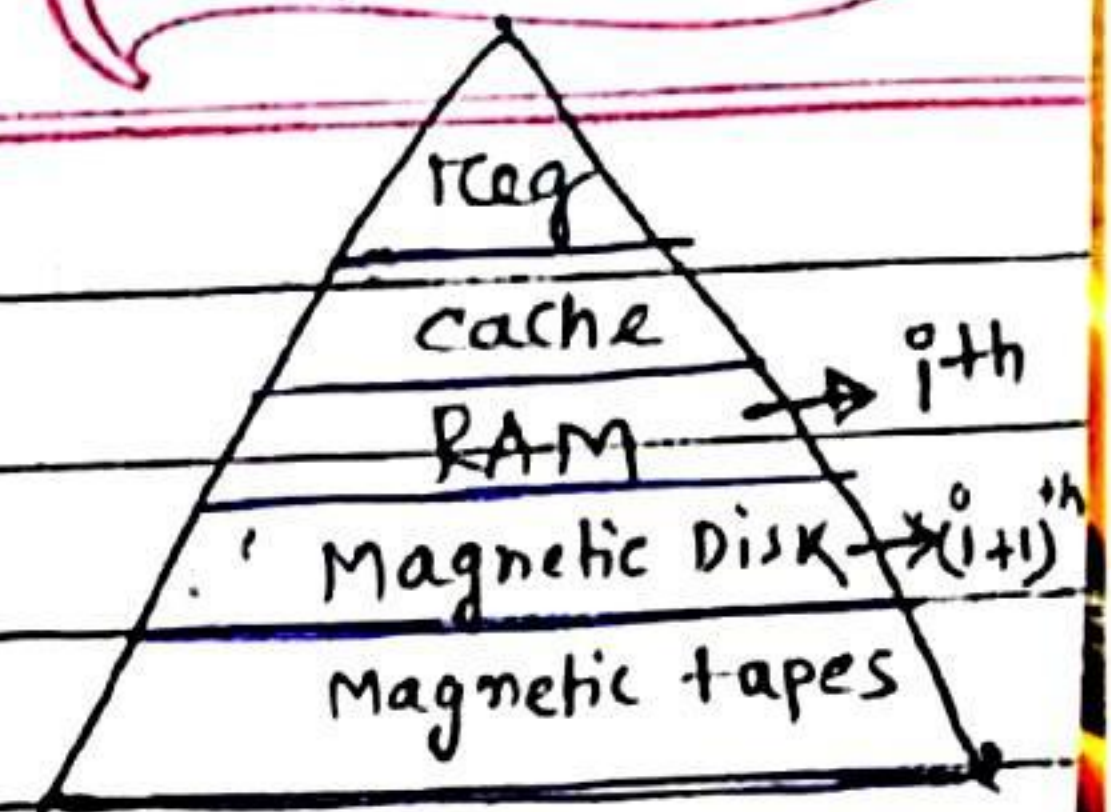
→ The purpose of memory hierarchy is to bridge the speed ~~mis~~ mismatch between fastest processor to slow memory at reasonable cost.

→ The goal of memory hierarchy is to minimize average access time of entire memory system.

• Level memory -

• 2 level memory -

Information in i^{th} $(i+1)^{\text{th}}$ level

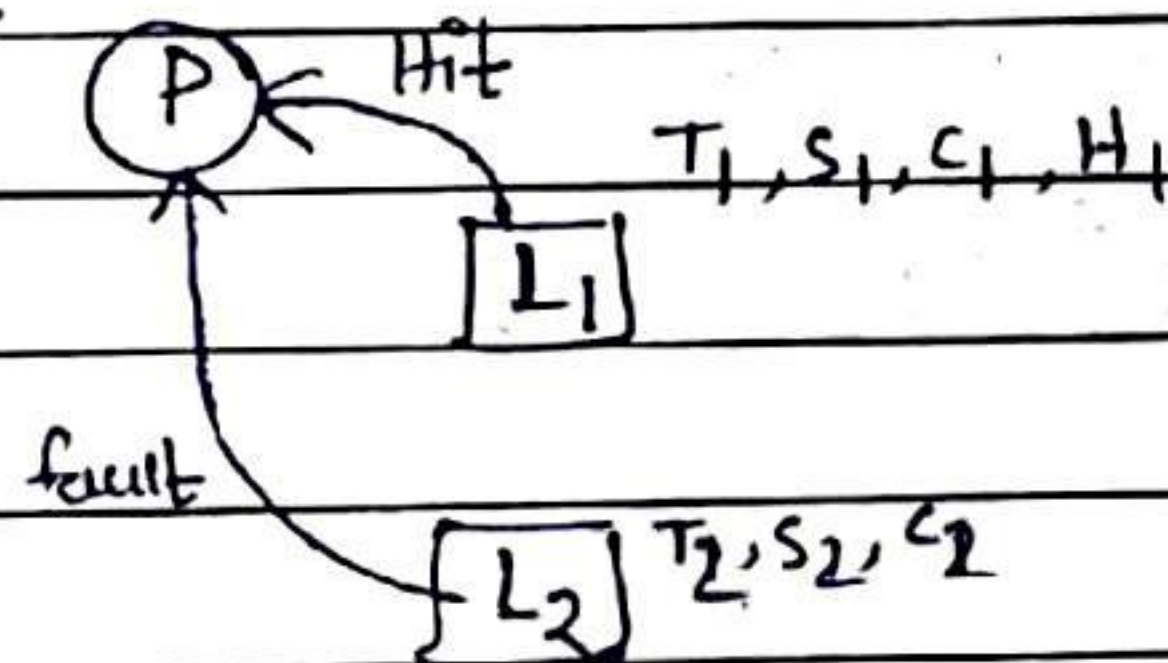


→ If processor refers to i^{th} level memory is found then "Hit" otherwise "Miss (or) fault".

→ There are two way in which the processor is connected to various levels of memory

Case 1:

Case



$$\text{Hit rate} = \frac{x}{100}$$

$$\text{Miss rate} = \left(1 - \frac{x}{100}\right)$$

$T_1 \rightarrow$ Time to access

$S_1 \rightarrow$ size of level 1 memory.

$C_1 \rightarrow$ cost per bit.

$H_1 \rightarrow$ Hit rate.

$$T_{\text{avg}} = H_1 T_1 + (1 - H_1) T_2$$

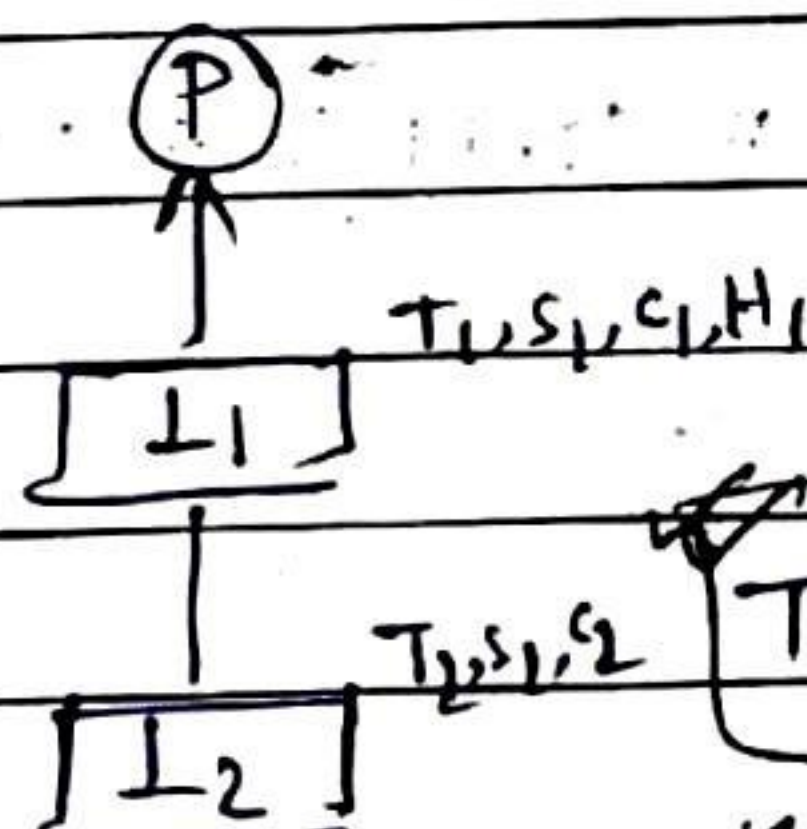
$$\text{Cost}_{\text{avg}} = \frac{C_1 S_1 + C_2 S_2}{S_1 + S_2}$$

100 $\rightarrow x$

$$T_{\text{avg}} = \frac{x T_1 + (100 - x) T_2}{100}$$

$$= H_1 T_1 + (1 - H_1) T_2$$

Case 2:



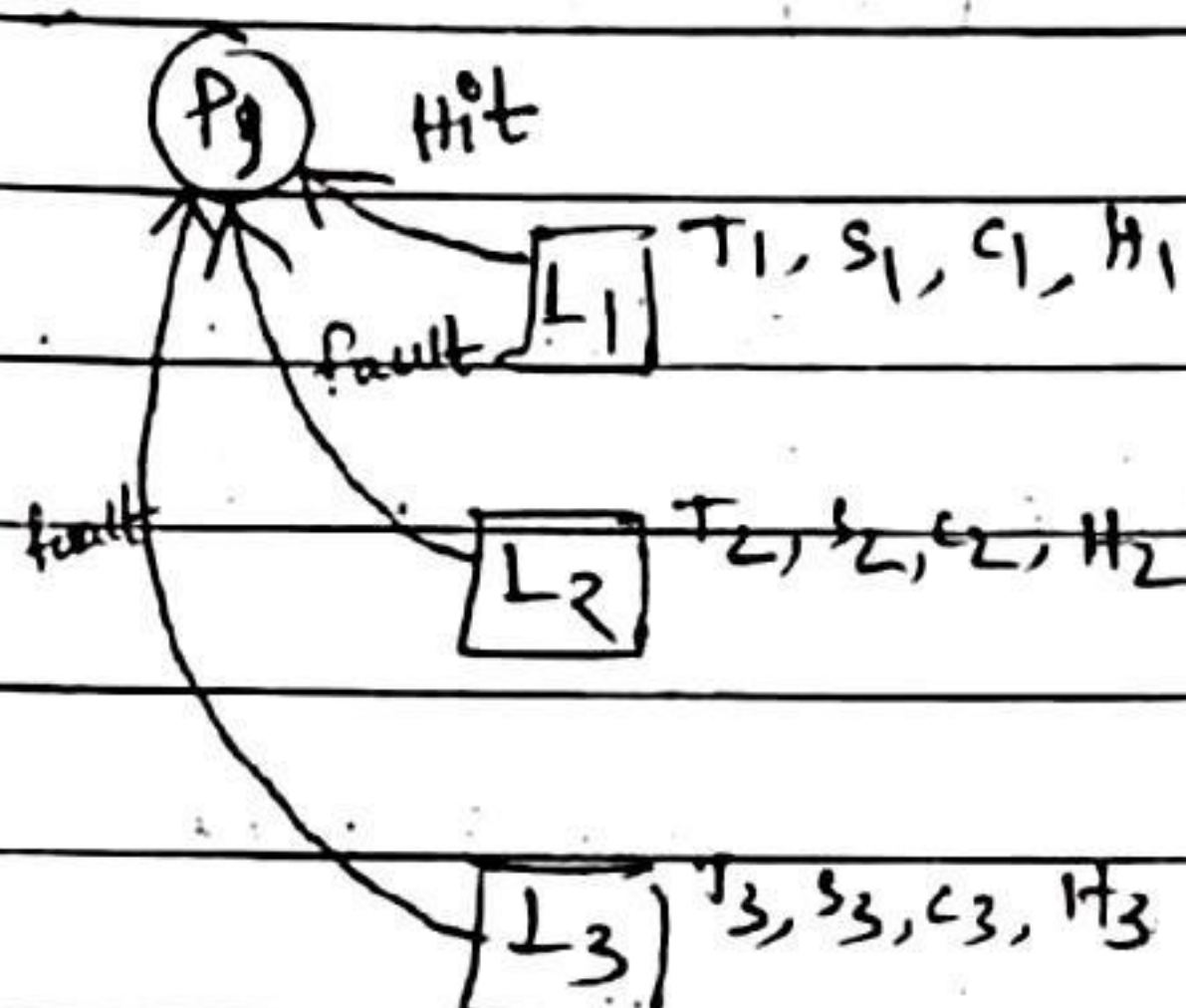
$$T_{\text{avg}} = H_1 T_1 + (1 - H_1) (T_1 + T_2)$$

$$\text{Cost}_{\text{avg}} = \frac{S_1 C_1 + S_2 C_2}{S_1 + S_2}$$

3-level memory -

case-1

$$T_1 < T_2 < T_3$$



Best case time

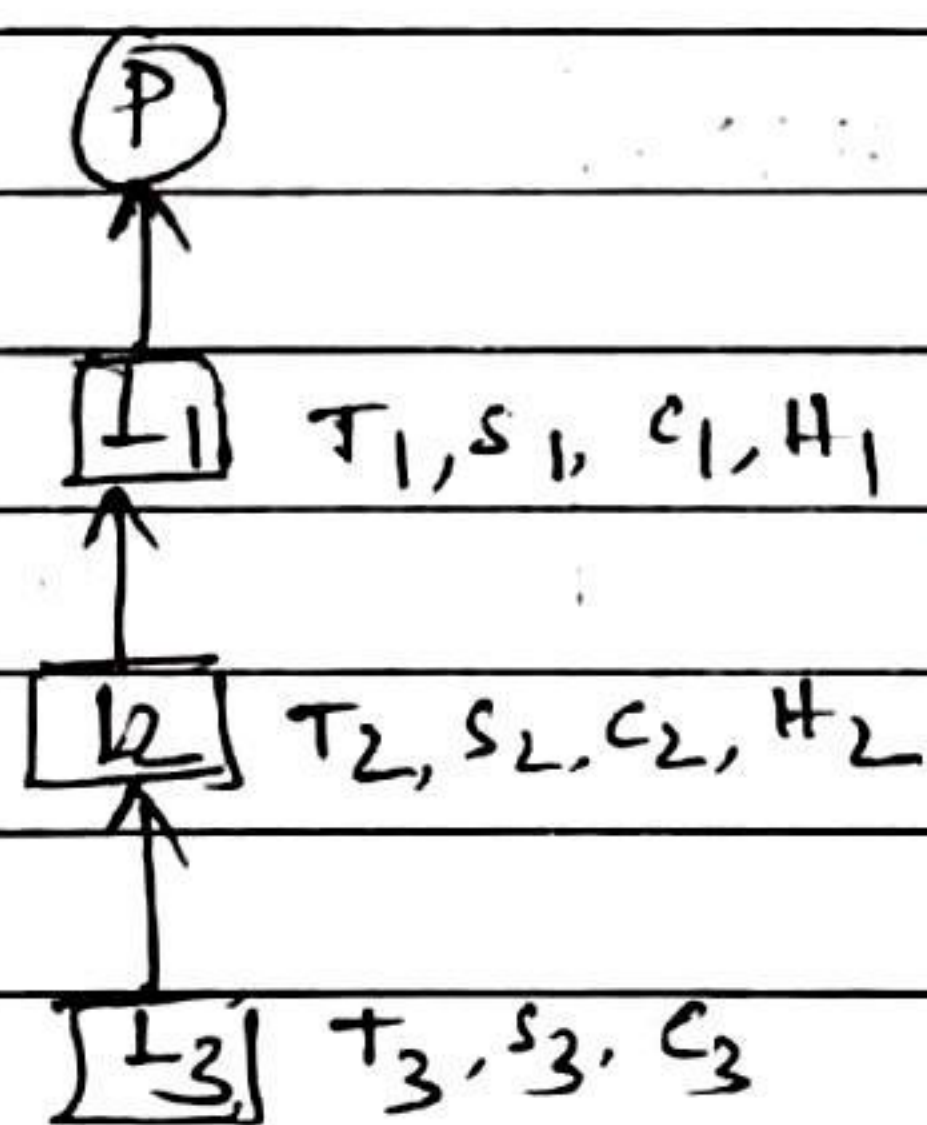
Worst time taken

$$T_1 \leq T_{avg} \leq T_3$$

$$T_{avg} = H_1 T_1 + (1-H_1) H_2 T_2 + (1-H_1)(1-H_2) T_3$$

$$C_{avg} = \frac{C_1 S_1 + C_2 S_2 + C_3 S_3}{S_1 + S_2 + S_3}$$

Case-2



Best time taken

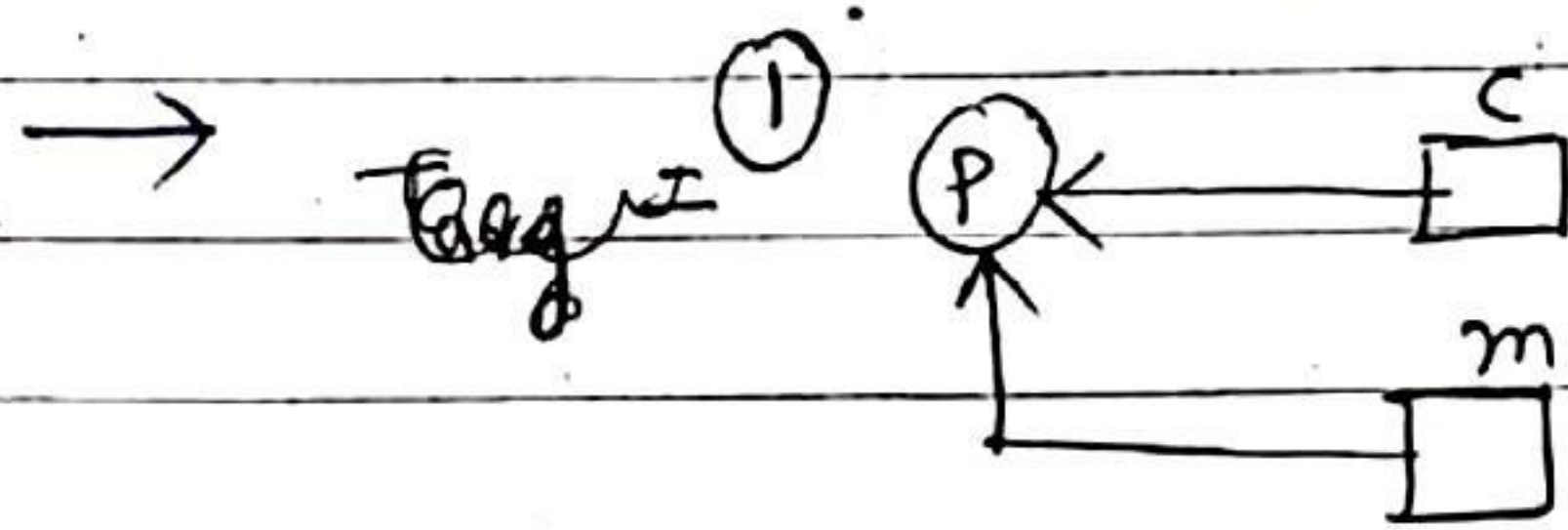
Worst time taken

$$T_1 \leq T_{avg} \leq (T_1 + T_2 + T_3)$$

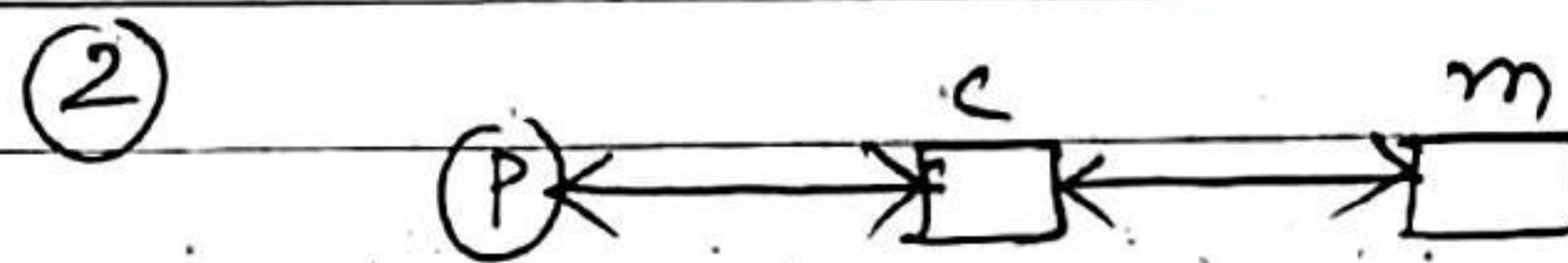
$$T_{avg} = H_1 T_1 + (1-H_1) H_2 (T_1 + T_2) + (1-H_1)(1-H_2) (T_1 + T_2 + T_3)$$

$$C_{avg} = \frac{S_1 C_1 + S_2 C_2 + S_3 C_3}{S_1 + S_2 + S_3}$$

Q: The average memory access time for a machine with a cache hit rate of 80% where the cache access time is 5ns and memory access time is 100ns is?



$$\begin{aligned}
 T_{avg} &= H_1 T_c + (1 - H_1) T_m \\
 &= (0.8)(5ns) + (1 - 0.8)(100ns) \\
 &= \boxed{24 ns}
 \end{aligned}$$



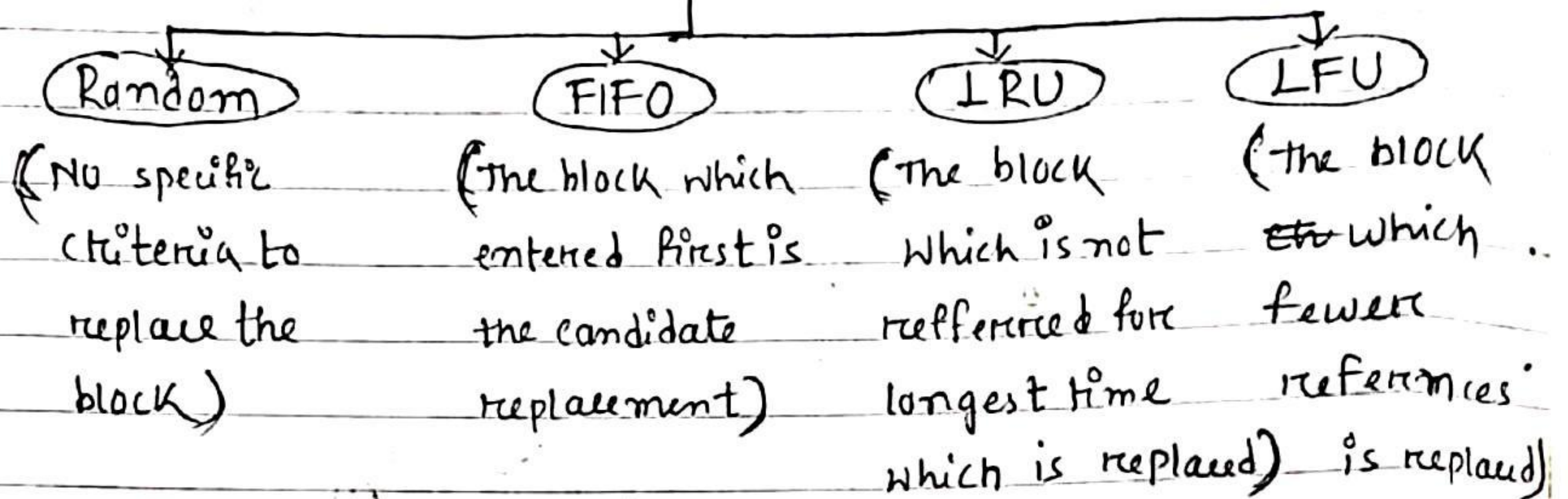
$$\begin{aligned}
 T_{avg} &= H_1 T_c + (1 - H_1) (T_c + T_m) \\
 &= (0.8)(5ns) + (1 - 0.8)(5ns + 100ns) \\
 &= 4ns + (0.2)(105ns) \\
 &= 4ns + 21.0ns \\
 &= \boxed{25 ns}
 \end{aligned}$$

• Cache replacement policy :-

→ Replacement policy is required for associative mapping and set associative mapping but not for direct mapping.

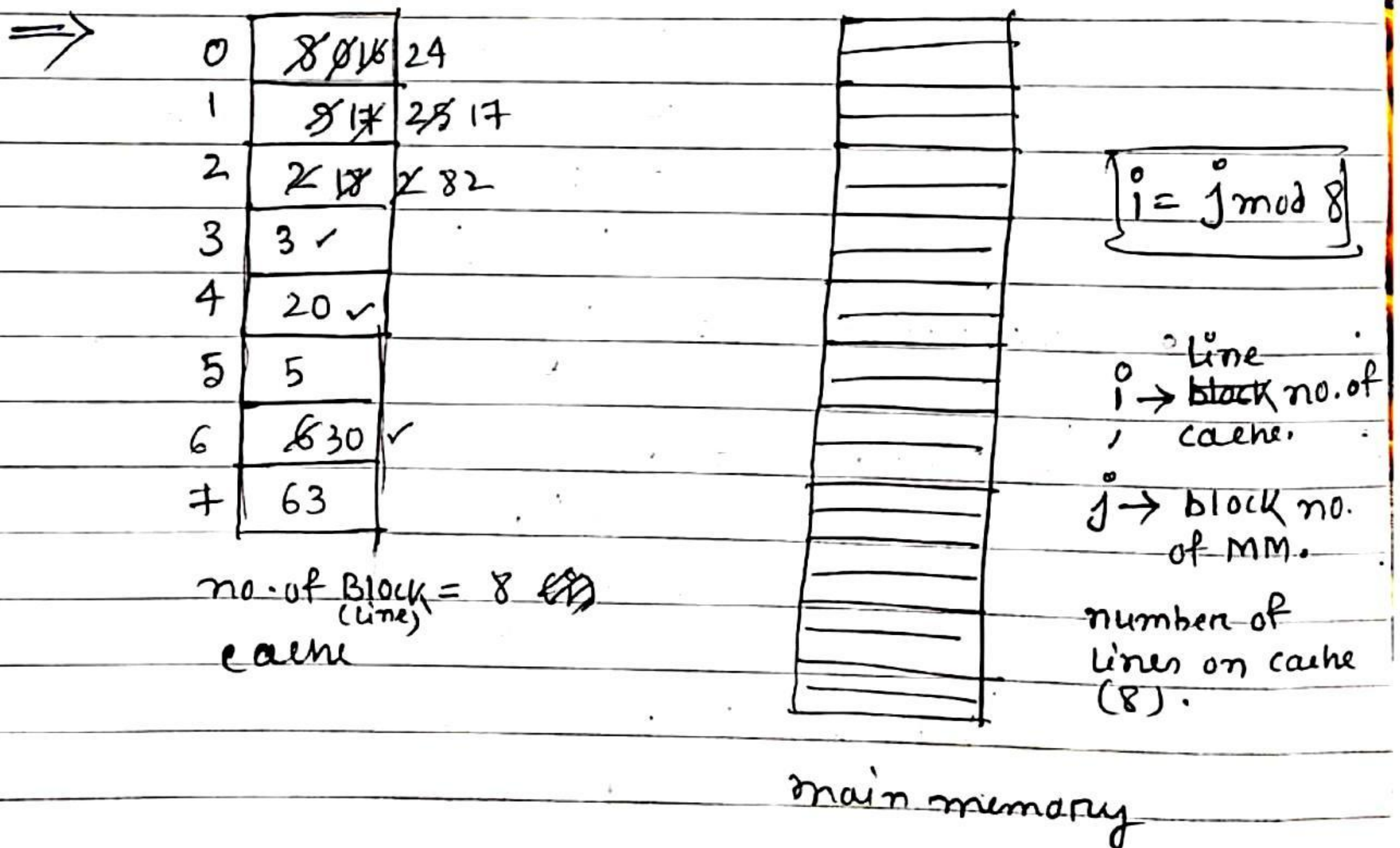
→ Replacement policies are aimed to minimize miss penalty for ~~references~~ future references.

Replacement Policies



Q: 1 consider a direct mapped cache with 8 cache blocks (0-7), if the memory block requests are in the order 3, 5, 2, 8, 0, 6, 3, 9, 16, 20, 17, 25, 18, 30, 29, 2, 63, 5, 82, 17, 24 which one of the memory blocks will ~~be~~ not be in the cache at the end of the sequence?

- (a) 3 (b) 18 (c) 20 (d) 30



Q: 2 Consider a 1-Way set associative mapping with 16 cache blocks, the more the memory block request are in the order (0, 255, 1, 1, 3, 8, 133, 159, 216, 219, 48, 32, 73, 92, 155) which one of the following memory block, will ~~not~~ be in the cache if LRU is used.

✓ 3

~~8~~

✓ 129

✓ 216

⇒

S ₀	0, 1, 8, 216 , 48, 32, 92 8, 48, 32, 92
S ₁	1, 133, 73,
S ₂	
S ₃	(255, 1, 155) , 219, 159 , 63, 155

(LRU - least recently used)

$$\text{Set-number} = (i) \bmod 4$$

i → Block number.

Cache memory (4-way set associative)

(means ~~one~~ each set contain 4 lines)

Q: 3 Consider a small 2-way set associative mapping with a total of 4 blocks, for choosing the block to be replace use LRU scheme, the number of cache misses for the following sequence of block addresses 8, 12, 0, 12, 8 is 4 ?

M M M H M

⇒

S ₀	8, 12 , 0 , 12, 8
S ₁	

cache

$$\text{s.no} = (i) \bmod 2$$

$$\checkmark \text{Miss rate} = \left(\frac{4}{5} \times 100\right) = 80\%$$

$$\checkmark \text{Hit rate} = \left(\frac{1}{5} \times 100\right) = 20\%$$

Q: 4 Consider a 2-way set associative mapping consisting of 2^c memory blocks and $2c$ cache blocks, the cache location for the memory block 'K' is —

- a) $K \bmod 2c$
 b) $K \bmod 2^c$
 c) $K \bmod c$
 d) $K \bmod K$



no. of lines in cache = $2c$

2-way set-associative.

no. of set = $\frac{2c}{2} = c$
~~set no = $K \bmod 2c$~~

Set no = $K \bmod c$

Q: 5

Consider the cache has 4 blocks, for the memory reference (5, 12, 13, 17, 4, 12, 13, 17, 2, 13, 19, 13, 43, 61, 19) what is the hit ratio for the following cache replacement algorithms —

- (i) FIFO (ii) LRU (iii) Direct mapping.
 (iv) 2-Way Set Associate (LRU):

→ (i) FIFO.

(5, 12, 13, 17, 4, 12, 13, 17, 2, 13, 19, 13, 43, 61, 19)

5	43
12	61
13	19
17	13

hit ratio = $\left(\frac{5}{15} \times 100\right)$

miss ratio = $\left(\frac{10}{15} \times 100\right)$

→ (ii) LRU. $(\overset{m}{5}, \overset{m}{12}, \overset{m}{13}, \overset{m}{17}, \overset{m}{4}, \overset{\checkmark}{12}, \overset{\checkmark}{13}, \overset{\checkmark}{17}, \overset{m}{2}, \overset{\checkmark}{13}, \overset{m}{19}, \overset{\checkmark}{13}, \overset{m}{43}, \overset{m}{61}, \overset{\checkmark}{19})$

$(\underline{5}, \underline{12}, \underline{13}, \underline{17}, \underline{4}, \underline{12}, \underline{13}, \underline{17}, \underline{2}, \underline{13}, \underline{19}, \underline{13}, \underline{43}, \underline{61}, \underline{19})$

$$\text{hit ratio} = \left(\frac{6}{15} \times 100\right)$$

$$\text{miss ratio} = \left(\frac{9}{15} \times 100\right)$$

(iii) → Direct mapping $(\overset{m}{5}, \overset{m}{12}, \overset{m}{13}, \overset{m}{17}, \overset{m}{4}, \overset{m}{12}, \overset{m}{13}, \overset{m}{17}, \overset{m}{2}, \overset{m}{13}, \overset{m}{19}, \overset{\checkmark}{13}, \overset{m}{43}, \overset{m}{61}, \overset{m}{19})$

0	12	17 2
1	5	13 17 13 17 61
2		
3	13	43 19

$$4(6) / 15$$

$$\text{line no} = (\text{B.NO}) \bmod 4$$

$$\text{cache hit ratio} = \left(\frac{1}{15} \times 100\right)$$

$$\text{miss ratio} = \left(\frac{14}{15} \times 100\right)$$

→ (iv) 2-Way Set Associate (LRU) -

$(\overset{m}{5}, \overset{m}{12}, \overset{m}{13}, \overset{m}{17}, \overset{m}{4}, \overset{m}{12}, \overset{m}{13}, \overset{m}{17}, \overset{\checkmark}{2}, \overset{m}{13}, \overset{\checkmark}{19}, \overset{\checkmark}{13}, \overset{\checkmark}{43}, \overset{m}{61}, \overset{m}{19})$

S ₀	12, 4 12, 2
S ₁	5, 13 17, 13, 17, 13, 13, 13, 43, 61, 19

$$\text{line no} = (i \bmod 2)$$

$i \rightarrow \text{Block no.}$

$$\text{hit ratio} = \left(\frac{5}{15} \times 100\right)$$

$$\text{miss ratio} = \left(\frac{10}{15} \times 100\right)$$

Q: 6

A hierarchical memory system has the following specification 20MB main storage with access time of 300ns, 256 B cache with access time of 50ns, word size 4B, page size 8 words. What will be the hit ratio if the page address trace of a program has the pattern 0, 1, 2, 3, 0, 1, 3, 0, 1, 2, 4 following LRU page replacement technique.



$$\text{Cache size} = 256 \text{ B}$$

$$\text{Word size} = 4 \text{ B}$$

$$\text{page size} = 8 \text{ Words} = (8 \times 4) \text{ B} = 32 \text{ B}$$

$$\text{No. of cache page} = \frac{\text{size of cache}}{\text{cache page size}} = \frac{256}{32} = 2^3$$

(0, 1, 2, 3, 0, 1, 3, 0, 1, 2, 4)

$$\text{hit ratio} = \left(\frac{3}{8} \times 100 \right)$$

Q: 7

consider an array A[100] and each element occupies 4 words, A 32-word cache is used and divided into 8-word blocks.

a) What is the hit ratio for the statement.

for (i = 0; i < 100; i++)

A[i] = A[i] + 10.

$$\rightarrow \text{Cache size} = 32 \text{ W}$$

$$\text{Block size} = 8 \text{ W}$$

$$\text{no. of Block in cache} = \frac{32}{8} = 4 \text{ Block.}$$

B ₀	A ₀	A ₁
B ₁	A ₂	A ₃
B ₂	A ₄	A ₅
B ₃	A ₆	A ₇

Cache.

$$\frac{A_0}{m}, \frac{A_0}{H}, \frac{A_1}{H}, \frac{A_1}{H}$$

$$\frac{A_2}{m}, \frac{A_2}{H}, \frac{A_3}{H}, \frac{A_3}{H}$$

hit rate = $\left(\frac{3}{4} \times 100\right) \checkmark = 75\%$ hit rate.

Q. 8

Consider an array has 100 elements and each elements occupies 4 words. A 32 bit word cache is used and divided into a block of 8 words. What is the hit rate of.

for (i = 0; i < 10; i++)
for (j = 0; j < 10; j++)

$$A[i][j] = A[i][j] + 10.$$



Cache size = 32 Word

Block size = 8 W

no. of Block = 4.

i = 0 10
j = 1
j = 2

B ₀	A ₀	A ₄
B ₁		
B ₂		
B ₃		

Cache.

array = [A₀, A₁, A₂, ..., A₁₀₀]

	RMO					
00	01	02	...	09		
10	11	12	...	19		
20						
30						
:						
90	91	92	...	99		

CMO

[00] [10] [20] [30]

PMO :

	00	01
00		
01		
10		
11		

m	H	H	H
00	00	01	01
R	W	R	W

$$TM = (50)1$$

$$TH = (50)43$$

$$TR_{reference} = (50)4$$

$$\text{Hit ratio} = \left(\frac{3}{4} \times 100 \right) \\ = 75\%$$

(column major order)

CMO :

m	H	m	H
00	00	01	01

$$TM = (50)2$$

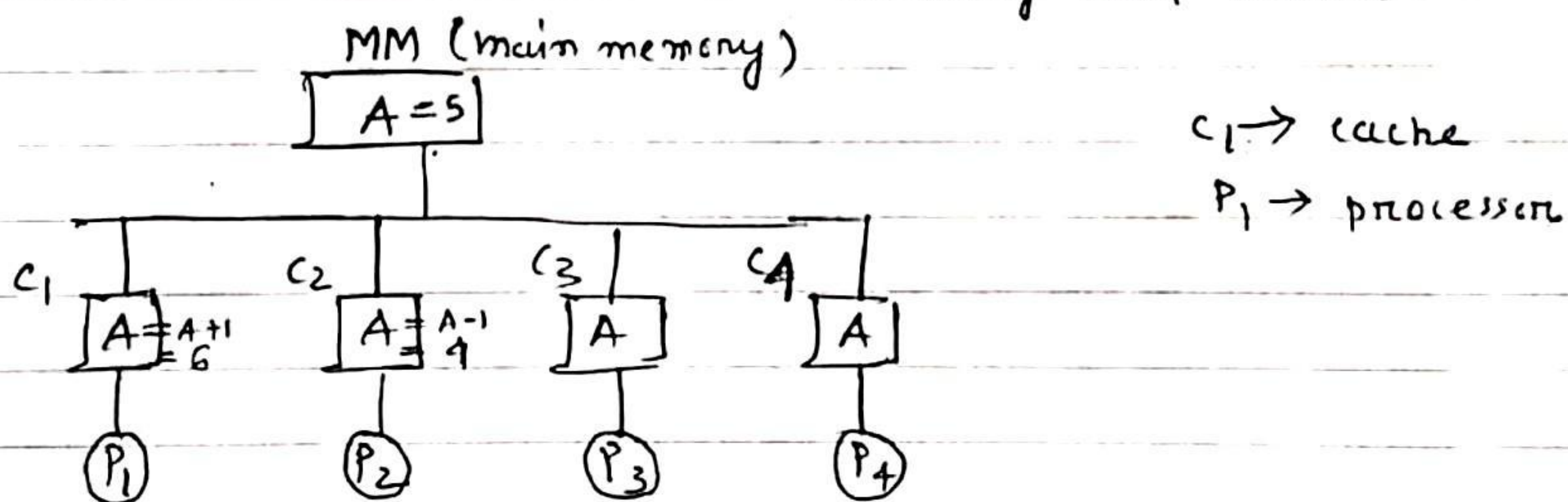
$$TH = (50)2$$

$$TR = (50)4$$

$$\text{Hit ratio} = \left(\frac{2}{4} \times 100 \right) \\ = 50\%$$

• Cache Coherence problem :

→ cache coherence problem : Multiple copies of same data can exist in different caches simultaneously, and if processors are allowed to update their own copies freely an ~~an~~ inconsistent view of memory can result.

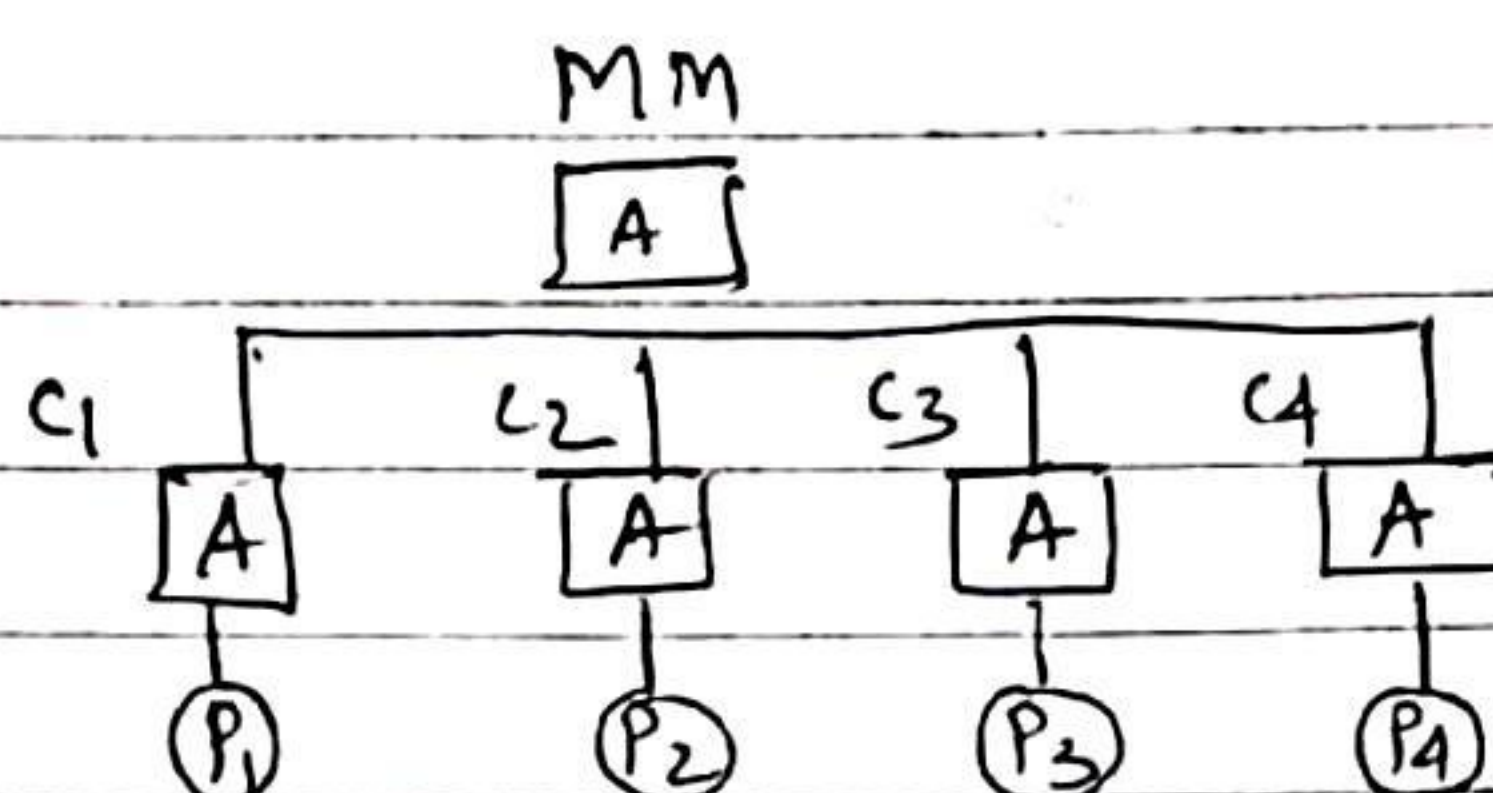


• Methods to avoid cache coherence problem -

→

There are 4-method to avoid cache coherence problem

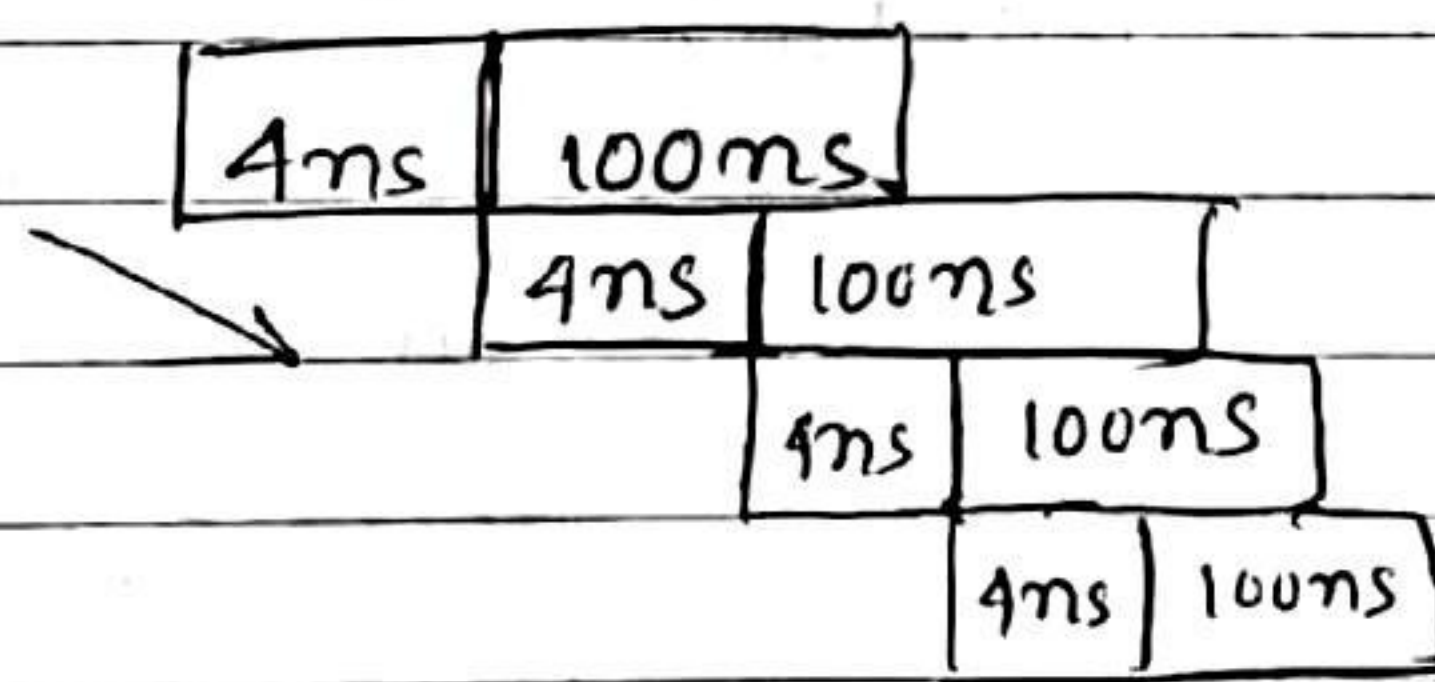
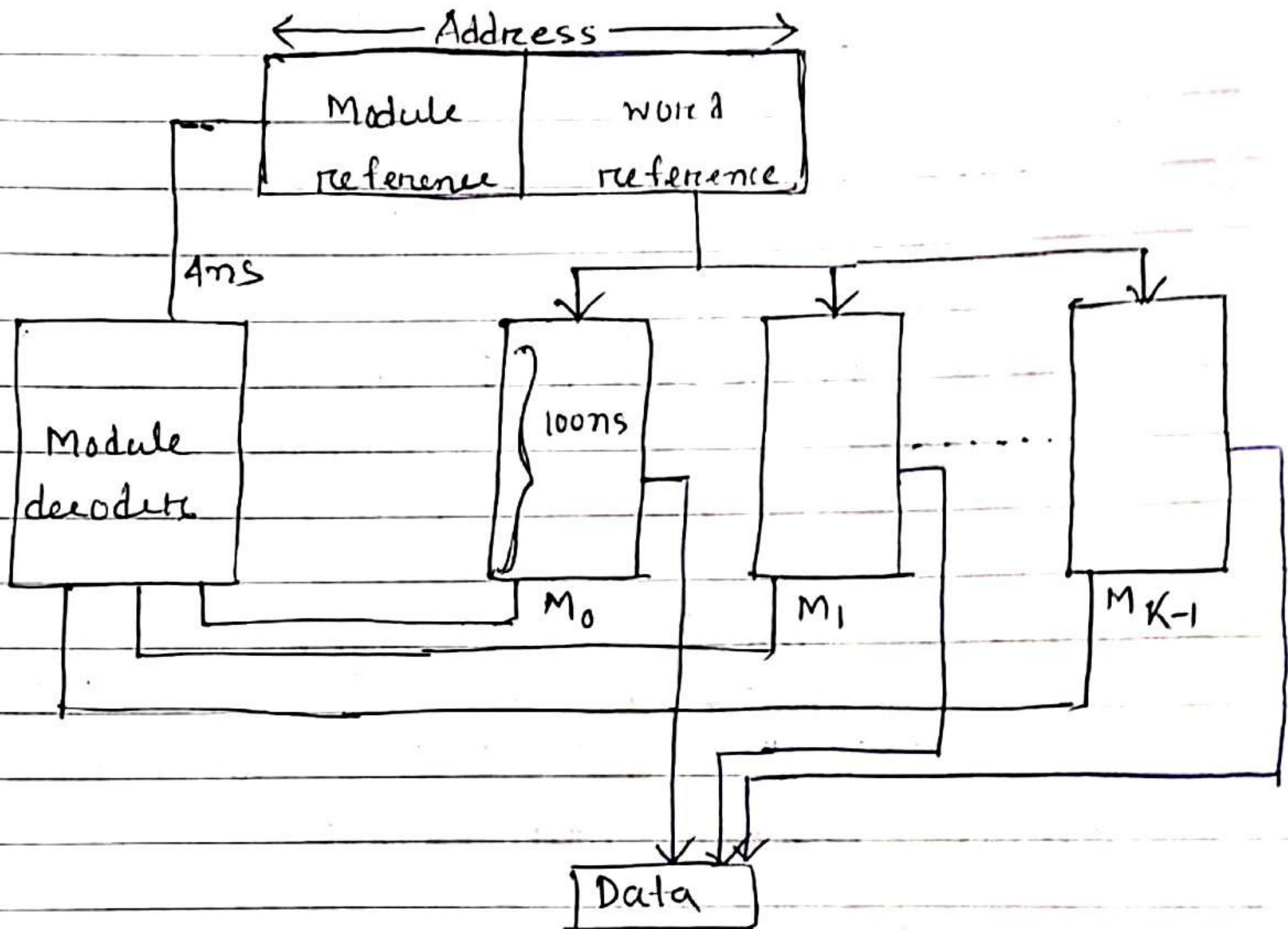
- 1) write Update → write through.
- 2) write Update → write back.
- 3) write invalidate → write through.
- 4) write invalidate → write back.



- 1) Write update - write through : update simultaneously of a word in cache & - M memory
- 2) Write update - ^{write} Back : The updation of MM is postponed until the associated block is replaced.
- 3) Write Invalidate - write through : ~~changing a value in~~ ^{the value} ~~update simultaneously of a value in one cache, will be invalid~~ ^{MM. processor and other can't use this value.} ~~ate to other cache.~~
- 4) Write Invalidate - write Back :
- 3) write invalidate - write through : if processor P_1 ~~changed~~ ^{changes} a value and change, then ~~other other~~ processor P_2 can't use the value will be invalid for other processor and update simultaneously of value in ~~mm~~ MM.
- 4) Write Invalidate - write Back : if P_1 ~~and~~ ^{change} a value then this value will be invalid to other processor and updation of MM is postponed until the associated block is replaced.

• Memory interleaving :

- "Reduce average access time".
- "Improve data transfer rate".



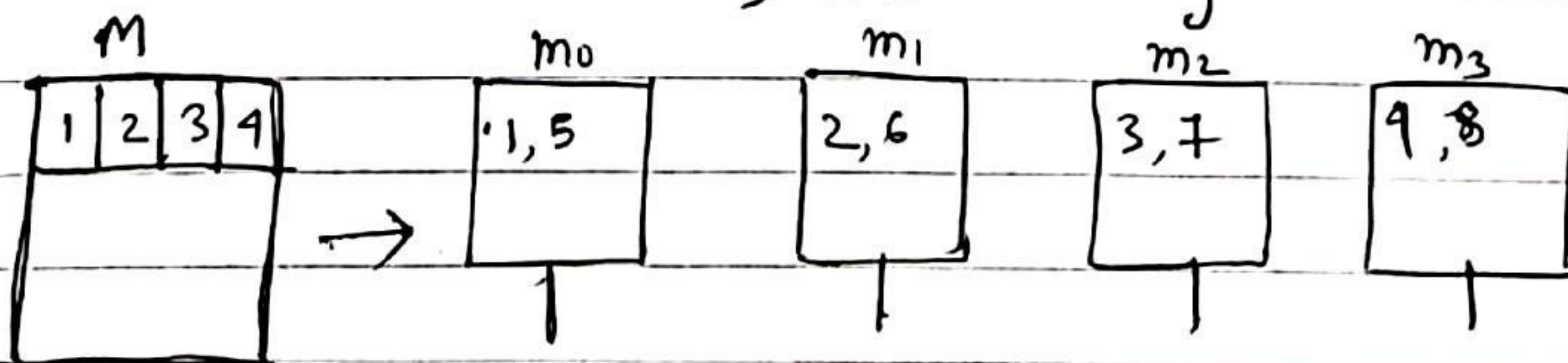
$$K(4ns) + 100ns$$

If read 'K' words then, total time taken, $T = K(4ns) + 100ns$.

AND

Without Interleaving concept total time, $T = K(100ns)$.

memory interleaving =



$$1 \text{ ms} = 10^{-3} \text{ s}$$

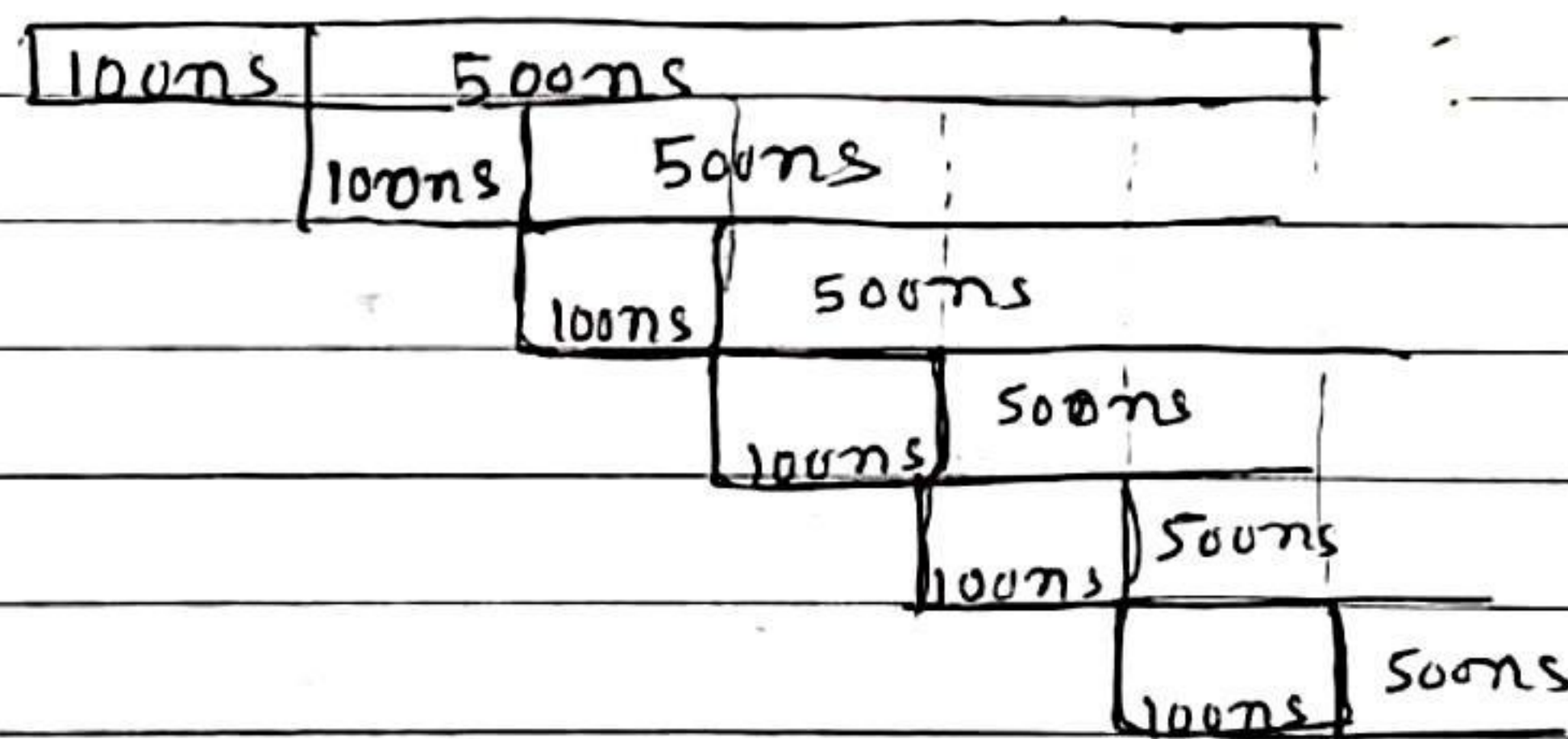
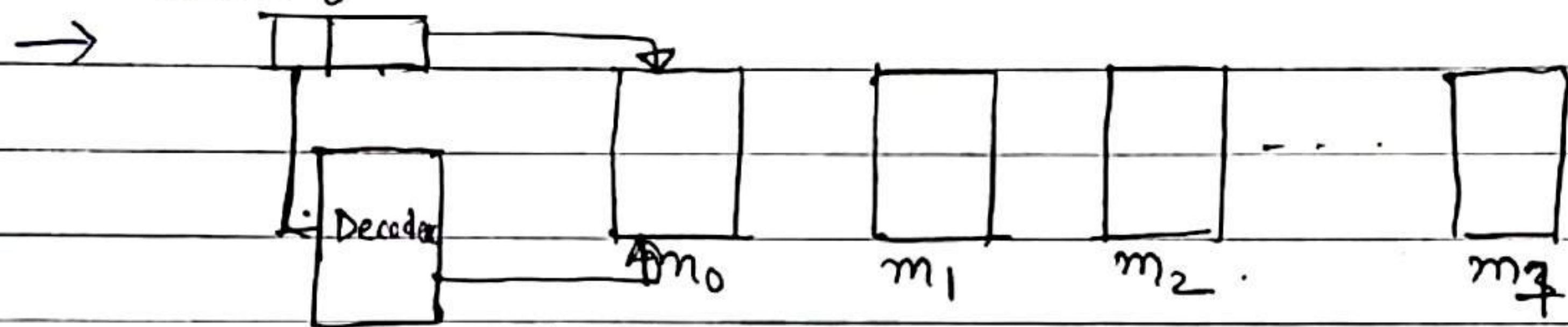
$$1 \text{ ns} = 10^{-9} \text{ s}$$

classmate

43

Q:1

Consider a memory system that contains 8 memory modules they are attached to the system bus which is one word length when a write request is made the bus is occupied for 100 ns with address data and control information, then after addressed memory module takes 500 ns for storing the word. If modules internal operations are overlapped what is will be the maximum numbers of request can be generated in 1 ms time?



100 ns — 1 request

$100 \times 10^{-9} \text{ s}$ — 1 request

1 s — $\frac{1}{100 \times 10^{-9}}$ request

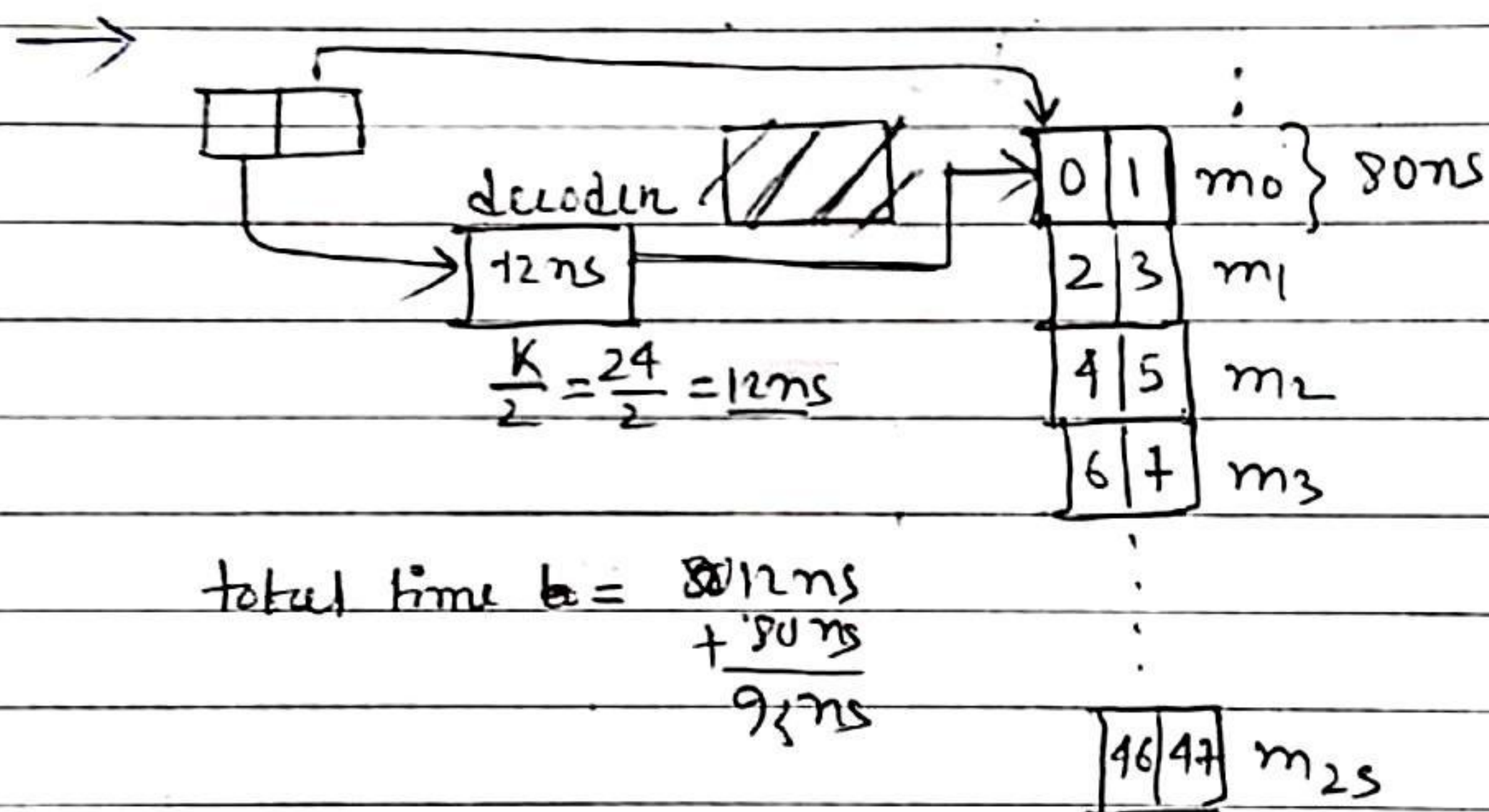
1 ms $\rightarrow \frac{10^{-3}}{100 \times 10^{-9}}$ request

$= 10^4 \text{ request} \checkmark$

Q. 1.1.1
N. Und. [5/12]

A CPU has a cache of 64 bytes. The main memory has K banks. Each bank can store C -bytes of data.

The ~~consecutive~~ consecutive ' C ' byte chunks are mapped to consecutive bank with wraparound manner, all K -banks can access in parallel. However, the two access for the same bank have to be serialised (one after the other). Let ' K ' equal to 24 and ' C ' equal to 2. Each iteration requires decoding the bank numbers to be accessed in parallel and this takes $\frac{K}{2}$ ns latency of each addressing bank is 80 ns. How much time is required to transfer initial block of cache?

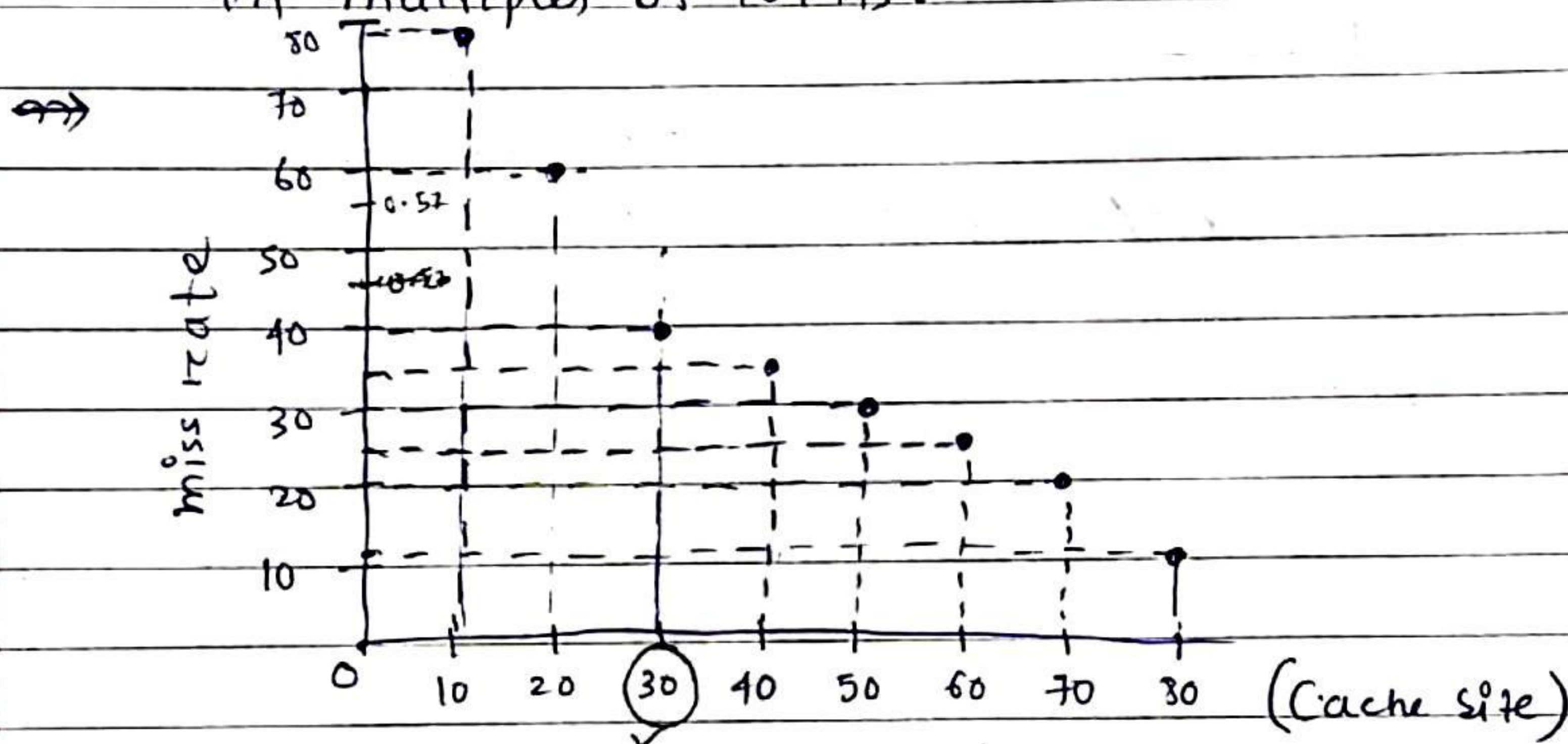


$$\text{total time } t = \begin{array}{r} 80 \text{ ns} \\ + 12 \text{ ns} \\ \hline 92 \text{ ns} \end{array}$$

$$= (92 \times 2) \text{ ns} \\ = 180 \text{ ns} \checkmark$$

9-2016

Q:3 A file system uses an in memory cache to cache disk blocks. The missrate of the cache is shown in the figure. The latency to read a block from the cache is 4ms and to read a block from disk is 10ms. Assume that the cost of checking whether a block exist in the cache is negligible. Available cache size are in multiples of 10MB.



The smallest cache size required to ensure an avg read latency less than 6ms is 30 MB?

→

Read latency < 6ms

 $(1-H_1)$
 so, miss rate < 0.57

$$H_1 T_1 + (1-H_1) T_2 < 6 \text{ ms}$$

$$H_1(4) + (1-H_1)10 < 6 \text{ ms}$$

$$\Rightarrow 4H_1 + 10 < 6 \text{ ms}$$

$$\Rightarrow 4H_1 < -4 \text{ ms}$$

~~$$H_1 < -1 \text{ ms}$$~~

$$4H_1 > 4 \text{ ms}$$

$$H_1 > \frac{4}{4} \text{ ms}$$

$$\Rightarrow H_1 > 0.433 \text{ ms}$$

$$\Rightarrow 1-H_1 < 0.433$$

$$\Rightarrow (1-H_1) < 1 - 0.433 \quad \Rightarrow 0.57$$

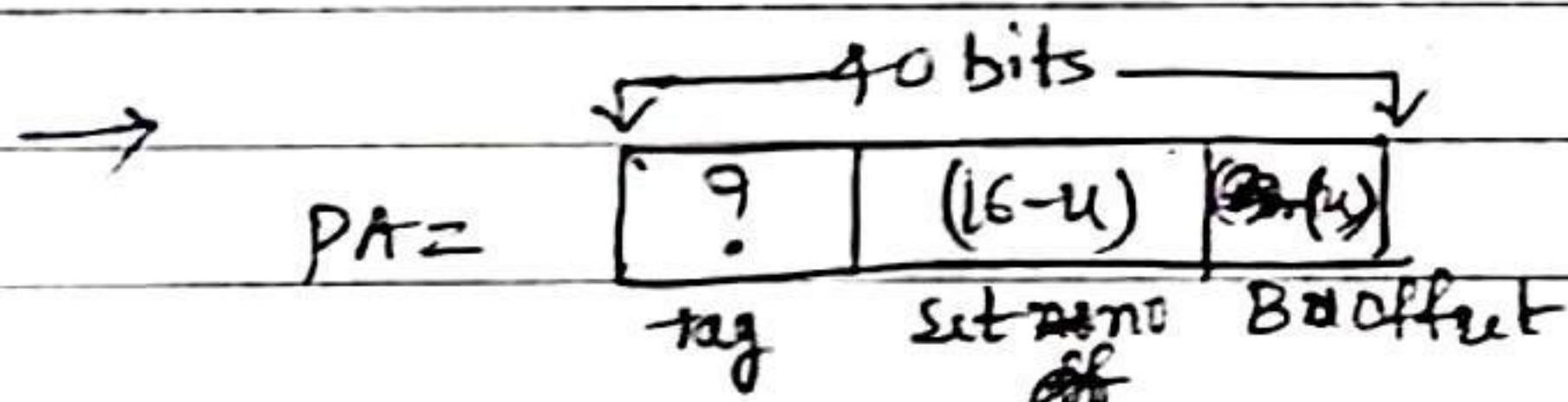
Q 14

A processor can support a maximum memory of 4 GB, where the memory is word addressable (a word consists of two bytes). The size of the address bus of the processor is at least 31 bits.

$$\begin{aligned} \rightarrow 4 \text{ GB} &= \frac{4 \times 2^{30} \text{ B}}{2} = 2^{31} \text{ B} & 4 \text{ GB} & \quad 1 \text{ W} \rightarrow 2 \text{ B} \\ & & \rightarrow (2^k \times 2 \text{ B}) \\ & & \rightarrow 2^k \text{ W} \\ & & \rightarrow 2^k \times 2^{30} \text{ W} \\ \text{total \# no. of blocks} & \rightarrow 2^{31} \text{ W} \end{aligned}$$

Q 15

The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is 16 bits.



$$\begin{aligned} \text{Cache size} &= 512 \text{ KB} \\ &= 2^{9+10} \text{ B} \\ &= 2^{19} \text{ B} \end{aligned}$$

$$\checkmark \text{ let, Block size} = 2^k$$

$$SS = 8 \text{ Block.}$$

$$\text{no. of blocks} = \frac{2^{19}}{2^k}$$

$$= 2^{(19-k)}$$

$$\checkmark \text{ no. of sets} = \frac{2^{19-k}}{2^3} = 2^{(16-k)}$$

$$\text{Tag bit} + (16-k) + (23-k) = 40$$

$$\Rightarrow \text{Tag bit} + 35 - 2k = 40$$

$$\Rightarrow \text{Tag bit} + 16 = 40$$

$$\boxed{\text{Tag bit} = 24}$$

9-2019

Q.6 The memory access time is 1 ns for a read operation with a hit in cache, 5 ns for a read operation with a miss in cache, 2 ns for ~~write~~ a write operation with a hit in cache and 10 ns for write a operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction ~~of~~ fetch operations. 60 memory read operations and 40 memory write operations. The cache hit ratio is 0.9. The avg memory access time (in ns) in executing the sequence of instructions is 1.68 ns?

→

$$T_R = (0.9)(1\text{ ns}) + (0.1)(5\text{ ns}) = 1.4\text{ ns}$$

$$T_W = (0.9)(2\text{ ns}) + (0.1)(10\text{ ns}) = 2.8\text{ ns}$$

$$(160)(1.4\text{ ns}) + (40)(2.8\text{ ns})$$

$$\Rightarrow (224 + 112)\text{ ns}$$

$$\Rightarrow 336\text{ ns}$$

$$\text{Avg memory time} = \frac{336}{\textcircled{200}}\text{ ns} \quad \text{--- total operation.}$$

$$= 1.68\text{ ns.}$$

9-2013

Q:7

Assume that for a certain processor, a read request takes 50ns on a cache miss and 5ns on a cache hit. Suppose while running a program, it was observed that 80% of processor read request in a cache hit. The avg. read access time (in ns) is 14 ns.

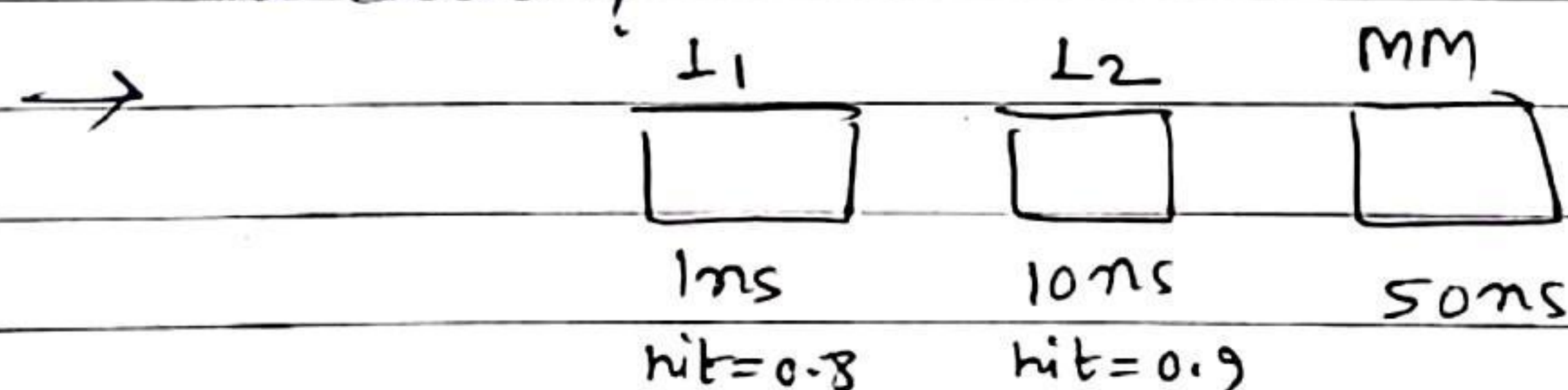
→ $T_R = (H_1)(5ns) + (1-H_1)(50ns)$

← avg access time

$$= (0.8)(5) + (0.2)(50)$$

$$= 4 + 10 = 14 \text{ ns}$$

Q:8 considers a system with 2-level cache. Access time of level-1, level-2 and main memory are 1ns, 10ns and 500ns respectively. The hit rates of level-1 and level-2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?



$$T_R = H_1 T_1 + (1-H_1) (H_2 T_2 + (1-H_2) T_3)$$

$$= (0.8)(1) + (1-0.8)(0.9)(10ns) + (1-0.8)(1-0.9)(500ns)$$

$$T_R = \boxed{12.6 \text{ ns}}$$

2006

classmate

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Q:9

A cache line is 64 bytes. The main memory has latency 32 ns and bandwidth 1 GB/s. The time required to fetch the entire cache line from the main memory is ?

→

Memory latency :

is the delay between the moment a memory controller tells the memory module to access a particular memory column and the moment the data is available on the module's output.

Bandwidth : The rate at which main memory can provide information.

→

1 GB/s 64 B

~~1 GB~~ -

1 GB - 1 s

1 B - $\frac{1}{10^9}$ s

64 B - 64 ns

10^9

64 ns + 32 ns

= 96 ns

→ Total time required.

9-1990

Q: 10 In a two level virtual memory ~~the memory~~ access time for main memory is 10^8 sec and the ~~the memory~~ access time for ~~the memory~~ the secondary memory is 10^3 sec. What must be the hit ratio, such that access efficiency within 80% of its maximum value?

$$\rightarrow (0.8)(10^{-3} \text{ sec}) = (H_1)(10^8) + (1-H_1)(10^3).$$

$$\Rightarrow 0.8 = (H_1)(10^5) + (1-H_1)$$

$$0.8 = H_1(10^5 - 1) + 1$$

$$H_1 = \frac{0.2}{10^5 - 1}$$

$$\rightarrow 0.2 = H_1(10^5 - 1)$$

$$H_1 = \frac{0.2}{1 - 10^5} = 0.200002 \approx 20\%$$

$$\frac{20}{100} = 0.20 \times 100\%$$